

and not substantially all claimed compounds exhibited a particular technical effect, the conclusion had to be that the invention as broadly defined in the independent claim was not a solution to the technical problem of achieving the given technical effect, with the consequence that the alleged technical effect of some of the claimed compounds was to be disregarded when determining the objective problem underlying the invention and thus when assessing inventive step.

In T. 942/98 the problem as per the application was the preparation of improved selective herbicides. With reference to T. 939/92 (OJ 1996, 309), the board did not agree with the applicant that, in the case in hand, he merely had to show an improved effect at the direct interface between prior art and the application in order to demonstrate, without providing further details, that such an improvement applied without qualification for the entire breadth of the claim. The appellant's position, in the board's view, amounted to allowing the applicant to decide how broad a claim could be, whether or not an improvement in the effect of all the compounds claimed was actually credible.

In T. 415/11 the board stated when the credibility that a technical effect is achieved by substantially all claimed compounds is at issue and in a situation where, it is prima facie unlikely that this is credible, it is not the opponent, but the patentee who has the burden of proving that the effect is achieved (referring to T. 939/92, OJ 1996, 309; T. 97/00).

In T. 41/16 the board observed that it was the boards' established case law that a technical effect on which inventive step was based had to be achieved over substantially the whole of the claimed range (see e.g. T. 939/92, OJ 1996, 309). However, that did not mean that every composition encompassed by a claim had to show an improvement over any composition comprised in the prior art, or even just over the one closest to the claim. What was required instead was that each composition according to the invention that differed from an equivalent composition in the closest prior art only in that it had the feature distinguishing the claim from the prior art showed the alleged improvement.

In T. 1296/13 the board stated that Art. 69 EPC and its Protocol did not provide a basis for excluding from a claim embodiments that fell within the wording of the claim (referring to T. 223/05).

#### 9.9.4 Intermediate products

In decision T. 22/82 (OJ 1982, 341) the board ruled that the preparation of new intermediates for a surprisingly advantageous complete process for the preparation of known and desired end products was inventive.

Again in T. 163/84 (OJ 1987, 301) intermediate chemical products were held to be patentable on the grounds that their further processing to the known end products involved an inventive step. The board however held that a new chemical intermediate did not become inventive merely because it was prepared in the course of an inventive multi-stage process and was further processed to a known end-product; there had to be other factors as well, such as that the process for preparing the new intermediate had enabled it to be

prepared for the first time and had done so inventively and other methods of preparing it had appeared to be ruled out.

In T.648/88 (OJ 1991, 292) the board disagreed with the view expressed in T.163/84, pursuing instead the line taken in T.22/82. An intermediate intended for the preparation of a known end-product was deemed to be inventive if its preparation took place in connection with inventive preparation or inventive further processing or in the course of an inventive complete process (confirmed in T.1239/01).

In T.65/82 (OJ 1983, 327) it was explained that new intermediates which take part in (non-inventive) analogy processes for sequent products (i.e. end products or intermediates of various kinds), must – in order to qualify as intermediates – provide a structural contribution to the subsequent products. Even where this condition was met, such intermediates were not thereby unconditionally inventive, i.e. not without taking the state of the art into consideration. As state of the art in relation to intermediates there are two different areas to be taken into account. One is the "close-to-the-intermediate" state of the art. These are all compounds identified from their chemical composition as lying close to the intermediates. On the other hand the "close-to-the-product" state of the art must also be taken into account, i.e. those compounds identified from their chemical composition as lying close to the subsequent products. With respect to the **"close-to-the intermediate" state of the art**, the question was whether or not the skilled man could have deduced from it the need to carry out certain purposive modifications to known compounds in order to obtain that intermediate which alone could enable him to solve the problem of making the subsequent products by means of a specific analogy process. With respect to the **"close-to-the product" state of the art**, a further question then was whether or not the skilled man could have derived from it the claimed intermediate in an obvious fashion.

In T.18/88 (OJ 1992, 107) the applicants had argued that the insecticidal activity of the known end products was significantly superior to that of another known insecticide with a similar structure; this was sufficient to establish an inventive step for the intermediate products, even if the end products were not novel and/or inventive. The board, referring to T.65/82 (OJ 1983, 327), rejected the applicants' argument on the following grounds: claimed intermediates must themselves be based on an inventive step to be patentable. Whether, under certain circumstances, new and inventive subsequent products might support an inventive step of intermediates was not the question here, because the subsequent products in this case were either not novel or not inventive. The superior effect of subsequent products which were neither novel nor inventive was not sufficient to render the intermediates inventive (T.697/96, T.51/98).

#### 9.9.5 Predictable improvements resulting from amorphous forms as compared to crystalline forms

In T.777/08 (OJ 2011, 633) the claims in question related to a particular polymorph (form IV) of crystalline atorvastatin hydrate. The board found that the skilled person in the field of pharmaceutical drug development would have been aware of the fact that instances of polymorphism were commonplace in molecules of interest to the pharmaceutical industry, and would have known it to be advisable to screen for polymorphs early on in the drug

development process. Moreover, he would be familiar with routine methods of screening. Consequently, in the absence of any technical prejudice and in the absence of any unexpected property, the mere provision of a crystalline form of a known pharmaceutically active compound could not be regarded as involving an inventive step (see also T. 41/17, T. 1831/18). Further, when starting from the amorphous form of a pharmaceutically active compound as closest prior art, the skilled person would have a clear expectation that a crystalline form thereof would provide a solution to the problem of providing a product having improved filterability and drying characteristics. The arbitrary selection of a specific polymorph from a group of equally suitable candidates cannot be viewed as involving an inventive step.

In T. 478/17 the board distinguished the case in hand from T. 777/08 and T. 41/07. The case in hand was not about the selection of any crystalline form but about the selection of one specific salt, namely eliglustat hemitartrate, in which at least 70% by weight of the salt was crystalline. The selection of this specific salt was not arbitrary. Rather, this salt had unexpected properties, namely an improved (reduced) hygroscopicity and an improved chemical stability. The board could not see any "one-way street situation" in view of D1, as claimed by the appellant. The skilled person starting from D1 would have had various choices in terms of stoichiometry and degree of crystallinity. It concluded that, having regard to the cited prior art, it would not have been obvious to the skilled person to isolate eliglustat hemitartrate in which at least 70% by weight of the salt was crystalline and arrive at the compound as defined in claim 19 of the main request.

In T. 1684/16 the board distinguished the case in hand from T. 777/08, in the light of which the appellant submitted that the solution in the case in hand was obvious since screening of polymorphs was a routine task as demonstrated in the prior art. Unlike T. 777/08 the case in hand was not about the selection of any crystalline form but about the selection of one specific crystalline form (Form I of bosutinib monohydrate). The board found that the selection of this specific crystalline form was not arbitrary, but rather this form had unexpected properties, namely an improved stability when compared with the other crystalline forms in D1, D2 and D3. The fact that the skilled person was taught in the prior art to investigate polymorphs in order to isolate the crystalline form having the most desirable properties was in itself not necessarily sufficient to consider a specific polymorphic form having a certain desired property obvious (see also T. 1326/18).

#### 9.9.6 Synergistic effects

In T. 1814/11 the problem to be solved was to provide an alternative synergistically active fungicidal composition based on prothioconazole. The board concluded that synergistic effects were not foreseeable, i.e. even if a combination of two specific compositions had a synergistic effect as in document 1, such synergy could not necessarily be expected if the structure of one of the two compositions were modified. **Synergy** was not in principle foreseeable and therefore could not be attributed to a specific mechanism of action and/or structure. The board dismissed the respondent's suggestion of **trial-and-error** experimentation as inappropriate in this case.

## 9.10. Equivalents

According to established case law of the boards of appeal, equivalents which are not disclosed in a published document must not be considered in assessing novelty, as this properly belongs to the examination for inventive step (see T 167/84, OJ 1987, 369; T 446/88; T 517/90; see also Guidelines G-VI, 2 – March 2022 version). In T 697/92 the board dealt with the concept of "equivalent means", according to which two means were equivalent if, despite having different embodiments, they fulfilled the same function with regard to the same result. Both means performed the same function if they shared the same basic idea, i.e. if they applied the same principle in the same way. The result was the totality of the technical effects produced by the means. In order to be considered as equivalents, the means had to achieve the same kind and quality of result. A means was thus not equivalent if, because of its different embodiment, it led to a result of the same kind but of a different quality or degree of effectiveness. The result did not necessarily even have to be better; it was sufficient for it to be different, since it was not the result itself which was patentable but the means by which it was achieved (see also T 818/93, T 929/02).

## 9.11. Selection inventions

In T 2623/11 the board stated that for selection inventions, only if the selection was connected to a particular technical effect, and if no hints existed which lead the skilled person to the selection, may an inventive step be accepted. In the case in hand, there was no support in the application for the claimed selection being connected to any particular technical effect. Moreover, hints existed in the state of the art which led the skilled person to the selection.

## 9.12. Problem inventions

The discovery of an unrecognised problem may in certain circumstances give rise to patentable subject-matter in spite of the fact that the claimed solution was retrospectively trivial and in itself obvious (see T 2/83, OJ 1984, 265; T 225/84). The posing of a new problem did not represent a contribution to the inventive merits of the solution if it could have been posed by the average person skilled in the art (T 109/82, OJ 1984, 473). It also had to be taken into consideration that it was the normal task of the skilled person to be constantly occupied with the elimination of deficiencies, the overcoming of drawbacks and the achievement of improvements of known devices and/or products (see T 15/81, OJ 1982, 2; T 195/84, OJ 1986, 121; T 455/91, OJ 1995, 684).

In T 532/88 the board confirmed the established principle that to address a problem simply by looking for ways of overcoming difficulties arising in the course of routine work did not constitute inventiveness. Following this case law, the boards held in T 630/92, T 798/92, T 578/92, T 610/95, T 805/97, T 1417/05 and T 2303/11 that the posing of the problem could not confer any inventive merit on the claimed subject-matter. Inventive step was however acknowledged in T 135/94, T 540/93, T 1236/03, T 764/12, T 1201/13 and T 2321/15 on the ground (also) that the posing of the problem was not obvious.

In T 252/10 the board noted that the absence of a hint in the prior art that there might still be a desire for further improvement did not mean that an unrecognised problem had been discovered.

In T 971/92 the board emphasised that the appreciation of conventional technical problems which formed the basis of the normal activities of the notional person skilled in the art, such as the removal of shortcomings, the optimisation of parameters or the saving of energy or time, could not involve an inventive step. The appreciation of a technical problem could thus only contribute to the inventive step in very exceptional circumstances. However, if an applicant nevertheless wished to rely on an assertion that the inventive activity resided in the recognition of a technical problem to which the solution was admittedly obvious, then the minimum requirement to be met was that this technical problem be clearly and unambiguously disclosed in the application as filed (see also T 43/97, T 1417/05).

In T 566/91 the invention related to a soft nystatin pastille formulation for treatment of candidiasis in the oral cavity. In the case in point the board did not agree with the submission by the appellants that the technical problem underlying the contested patent consisted in the unrecognised problem of poor patient compliance, as the average skilled person could have posed that problem where – as in that particular case – one necessarily came to light when an object or product was used. Consequently, a problem which amounted to no more than noticing obvious non-compliance with an obvious desideratum in a given situation, namely poor patient compliance using nystatin formulation as a result of the unpleasant taste of the active substance, could not be retained as the actual problem to be solved.

In T 764/12 the technical contribution of the patent in suit resided in identifying a problem which was hitherto not recognised in the prior art, namely the need for protection during storage at ambient environmental conditions of a chewing gum base comprising environmentally degradable polymers. The arguments of the opponent that the claimed invention lacked inventive step because the coating was well known to reduce degradation over time were not relevant. Referring to T 2/83 (OJ 1984, 265) the board acknowledged an inventive step because the perception of the problem had to be considered as being the main contribution to the inventive merits of the solution claimed.

### 9.13. New use of a known measure

When determining inventive step in the case of a new use of a known measure, the boards of appeal examine whether or not the problem which has been solved with a known measure in a known case differs from the problem posed in the case to be decided. If this examination reveals that there is no fundamental difference between the two problems, it can in principle be concluded that there is no inventive step if the known measure is adopted (see in particular T 39/82, OJ 1982, 419; T 142/84, OJ 1987, 112; T 332/90; T 485/91; T 25/97). In T 39/82 (OJ 1982, 419) the board stated that it could not be considered obvious for the skilled person to use a known measure in a different context since the problems differed fundamentally from one another.